CSC 648/848 Software Engineering – Spring 2026
Milestone 0

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Acknowledge
Collaboration and team project content is based on class material created by Dr. D. Petkovic & Anthony Souza

1.Introduction

The high-level purpose of this milestone is for student teams to choose, install and prepare IT infrastructure for the development of team final project in CSC 648-848. Specifically, in M0 student teams must achieve the following objectives working together:

1. Configuring **GitHub** to enable team SW development.
2. Choosing a **SW stack and delivery platform** and service provider from the given list
3. Getting your choices for SW stack and delivery platform approved by Class CTO
4. Creating a joint **WWW page** with info about team members using above infrastructure (this page will be graded but will also serve as ABOUT page for your team application)

The deployment server and ***all*** team applications for the final team project must be running on a remote host. ***Localhost hosting will not be accepted.***

In this class we will have **class CTO** who will deal with high-level technical issues and **class CEO** who deals with schedules etc. We also may have class TA to help class CTO and student teams with advice or use the **Team Project Technical Issues and Insights Exchange** forum in Canvas.

The work on M0 shall commence only after the teams have been formed and basic accounts set up – you will get e-mail on this from Class CTO

Note that your team will be choosing most of the IT infrastructure for developing the team project from the list provided in Milestone 0 – then the team choices will have to be approved by class CTO. After choosing the infrastructure and after it is approved all team members have to follow M0 and learn how to use the chosen tools. M0 and class IT involves infrastructure designed to give teams more freedom but more responsibility in maintaining your infrastructure. Hence, there may be problems. Please contact class CTO or class TA if you encounter any problems that you cannot solve yourself or with your team.

Your responsibility:

Since CSC 648-848 is a capstone senior course, students are expected to learn necessary tools themselves and by working with their team members. Use of on-line resources and asking around (team members, other classmates) is encouraged. Class instructors will only help in case there are problems, and only at high level. Also, specific coding details will be the duty of students to figure out with the help of on-line resources and asking around. Class instructors will also post some relevant resources (use of git/hub in team setting, several code templates for backbone infrastructure of the team project etc.).

The IT environment and setup we envision for the class is close to what you will be required to know to be a successful SW engineer, so M0 is a great experience to improve your marketable skills. While some of the work is to be done *individually*, you must work together with your team to complete the task. Note that your learning will consist of a) individual usage of tools; and b) most important: how to use the tools in team setting (e.g. code merge etc.).

Your choices of IT stack and hosting platform will be from a limited list: this is how it is in most companies, and it also simplifies our supervision and support to you.

M0 carries 10 grade points. Information about grading can be found at the end of the document. Again, each team member must complete his/her M0 tasks but to accomplish this each team member must also work with the rest of the team. **This means that you can help each other, ask for help, work in pairs and are in fact encouraged to do this to not only help with M0 but also build the teamwork.**

Your first task is to read through the entire document. Then, follow the steps starting with Section 2 "Setting up GitHub". You will also give access to instructors, so they can monitor your work.

In section 3 we then describe the actual M0 task where you are first to select deployment server and SW stack, then install them, and finally create a joint team web page. This WWW page can be used in your final project too (e.g. ABOUT page on your final team project WWW site). Section 4 describes M0 grading.

Again, M0 is a great vehicle to help you learn a very important part of SW engineering work which is setup, use and maintenance of professional development and deployment infrastructure.

Note that we use *group* or *team* interchangeably.

Please read this document carefully and follow all the required steps.

You must deliver M0 on schedule. Any delays must be approved by class CTO (Class CEO, CC'd) and asked for via e-mail at latest 24 h before the M0 deadline.

2. Setting up GitHub for team SW development

2.1 GitHub Team (Private) Repo

The purpose of this part of the exercise is for your team to set up the team private GitHub repository that is going to be used for storing your team's project (SW, documentation, formal milestone deliverables etc.) and be accessible only to the team members and instructors. Only one member needs to set up the private repository.

As a first step, all team members need to have their own GitHub account. This is mandatory, NO EXCEPTIONS - you need to have your own GitHub these days anyway.

Creating GitHub Team private repo.

1. Select one team member to create the private repo. Selected team member then uses the following link to create the private repo: https://classroom.github.com/a/-z9LTYpp

2. A chosen team member (team leads must ensure this happens) will then ADD ALL MEMBERS of the team to the private repo. Each invited team member has to accept the invite. **(For each non-invited team member, 1 point will be deducted from M0 Grade for the whole team to promote teamwork.)**
    a. **Simply inviting the team member is not enough**. They need to accept the invite. Non-confirmed invites will get the same penalty as no invite for those who do not follow up. **Team leads have to make sure team members respond to this**

Team members are strongly **encouraged to** practice creation of branches and code merge, which turned out to be occurring problems with previous teams. Please consult on-line resources for **GitHub** best practices on this topic and the slides we will post.

The team repository created with the above link will be used to store your team's source code and milestone documents. This repo is already private and configured for you to begin work right away.

<u>Instructors are already attached to the repo so you do not need to invite them</u>. Only your team members need to be added.

There is *<u>NO</u>* need for a paid subscription for the private repo. This has been taken care of for you.

Appoint GitHub master

Teams are also requested to appoint a GitHub master who will help organize and maintain GitHub according to best practices – M0 is a good time to do this.

3. M0 tasks

This section describes how to develop actual M0 tasks, after you have completed all the previous steps.

Task 1: Selecting Server/Platform Provider, Software stack and Deploying Your Team Web Application

The purpose of this task is for you as a team to do the following:

- Select a *Server/Platform provider*
  Some possible and strongly recommended choices are below:
  - Amazon AWS
  - Google Compute Engine
  - (Please refrain from using ***Heroku***. While simple, it has caused issues for many teams in the past that have delayed application development that has affected teams' final grade.)
- Select *a Software Stack*. An example would be a LAMP Stack. This includes a) the OS choice; b) web server choice; c) database choice; d) and server-side language choice, and e) any other important technologies needed your Software Stack.
- The server-side language for your web application must be one of the following:
  a. Python
  b. JavaScript/JSON
  c. PHP
  You may ***NOT*** use another server-side language that is not listed above. If you strongly disagree you may contact Class CEO and CTO with your suggestion.

IMPORTANT: When using AWS or Google Compute Engine please refrain from using their app features. These auto-deployment tools cause some issues for students and in some cases caused students to be charged (Most of it not all were refunded). If you do use them, use at your own risk we are not responsible for accrued costs for running the server.

For the DB we only allow MySQL for several reasons (well supported, standard SQL, available setup and management tools including Workbench, and has facilities we will use for search like %Like etc.)

When deciding on the technology to be used in your Software Stack, besides functionality (a common factor to start from), you should keep a few additional factors in mind when comparing technologies side by side. Some key factors may be:

- How often is the technology updated?
- How well is the technology supported?
- How mature is the technology?
- How well is the documentation?
- How good is the community using this tool?
- What features do is offer compare to other technologies.

- And very important: How easy is it to use/learn for all your team members given specific class schedules?
- Please avoid choosing the technologies where there is no expertise in the team. Remember: your focus is to deliver the application and NOT necessarily learn/use newest technologies

Note: your choices must be made such as not to incur any costs, which is possible today given many free offerings on the market. Appoint somebody to regularly (twice a week at least) checks cloud server billing and if there are issues act on it and/or talk to class CTO)

Task 2: Getting Server/Platform and Software Stack from Task 1 Approved:

The task of selection must be done very soon after formal start of M0 (e.g. within a week).

When your team has decided on a software stack, **team lead** needs to email the class CTO (cc'ing the class CEO as well) detailing your software stack. The email **must** have the following format:

- Receivers: Class CTO and CEO
- E-mail Subject line: CSC 648 848 Spring 2026 Section N Team M
  - **(N is section 02; M is team number)**
- List the Technologies used in your software stack.
- This includes the following:
  - Server Host, Instance size (CPU and RAM)
  - Operating System and Version Number
  - Database and Version Number
    - ***Only MySQL Database is allowed***
  - Webserver and Version Number
  - Server-Side language and Version Number
  - Also list any technologies or packages you will need that you think is important. (you can exclude things like git and SSH). There is no wrong answer here, just list what you can. The more the better, it'll help me determine how sound your software stack is.
- Then list each member's familiarity with the backend language. A sample is given in the Appendix of this document.

MAKE SURE TO FOLLOW EMAIL FORMAT. MOST IMPORTANTLY USE THE CORRECT SUBJECT HEADING. ANY EMAILS SENT WITH WRONG SUBJECT HEADING WILL BE EITHER RETURNED OR MISSED (IGNORED).

A sample email has been given in the Appendix of this document. Please do your best to follow this format. This will ensure a speedy response. If the class CTO has any questions about your software stack, make sure that your team replies promptly. Delayed responses can delay Software Stack approval.

Once your software stack has been approved via e-mail from class instructor you must begin installing and configuring your server and completion of M0 immediately.

Task 3: Installing and Configuring Approved Software Stack <u>(to be done after the above approval by class CTO).</u>

After your server/platform provider and software stack have been approved by e-mail from class instructor it is now time to begin installing everything.

There will be no detailed instructions given for this as there are too many possible configurations. But you may follow these simple steps:

1. Start server Instance with your server host
2. SSH/log into your server
3. Update your server
4. Install Webserver
5. Install DB
6. Install Server-Side language
7. Install remaining needed packages
8. Make any needed configurations.

Some notable Stack installation instructions:

- AWS w/ Node
- LAMP Stack AWS EC2
- LAMP Google Cloud Platform

This may not be the most detailed set of instructions, but it gives you an order of items to work through. **Please use on-line resources and follow the documentation!** If you run into any issues, there is help available. You may ask anyone in your team, your class, and class CTO.

THE CLASS CTO MUST BE GIVEN ROOT ACCESS TO THE TEAM SERVER (SSH ACCOUNT) AND DATABASE (DB USER WITH FULL PRIVELEGES). THERE IS NO EXCEPTION TO THIS.

Task 4: Correctly Populate Credentials Folder in Git Repository

The purpose of this task if for your team convey all the needed information for logging into your team's server and database. Failure to correctly give the required information will cause points to be deducted from your team's M0 grade as well as other milestones as well.

PLEASE DO NOT DO THIS TASK UNTIL TASK 3 IS COMPLETED.

In your team's repository there is a folder named credentials. The folder contains a README.md file with a set of instructions and required information. Please read this carefully and follow ALL instructions give ALL required information.
You may store all the required information in the README.md itself EXCEPT for SSH keys. These should be stored as files inside the credentials folder in your team's repository.

Task 5: Create a Team Website and ABOUT page

The purpose of this part of the exercise is to get you to work individually to create your own webpage within your team's framework context, and then to work with your team to join these pages together using your teams GitHub account and chosen framework into a single site. We recommend that you in fact create ABOUT WWW page which introduces the team members. This can then be part of your final application and is great for your portfolio.

Everyone in your team should clone the team's private GitHub repository into his/her individual shell account or onto your local computer, and within the chosen team framework create their own WWW page that at a minimum displays their name and their or some other image (if you are comfortable it is good idea to use your own photo which is useful for your ABOUT page and portfolio. **Photo is optional**). Please make sure that you have the right to use any posted photo. This work must be completed by individual student and then pushed to the team's private GitHub repository. The file(s) created/added should then be merged into the team's private GitHub repository.

The website must be deployed onto the team's remote server setup as in Task 3. ABOUT page and all student individual pages created for M0 task must also be deployed on the server.

Make sure your individual test environment is identical to the one on the server, see end of this section.
One of your teammates will need to modify your framework's home page to point to all the different team member pages (all residing at the server). The finished site must look something reasonably close to the example in Figure 1. The user 1...user 6 buttons should be replaced with buttons pointing to the individual pages labeled with the username of the person who created that page. **The website (home and individual pages) shall be deployed onto the team's group shell account**.

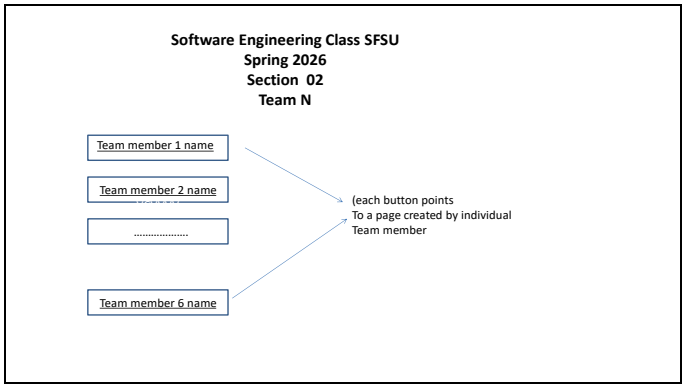


Figure 1. Example mockup of Milestone 0 team home page.

This single ABOUT page (and individual team member pages) site must **be served from the remote server you setup in Task 3.** This is the same way you will deploy your final app, hence this is useful to make sure you learn and test it, especially how to deal with branching and merging code. Every team must understand how your team's application is deployed and managed.

NOTE on student individual test environment:
Every student should have a correct test environment on their working PC e.g. **make SURE the same steps you do to configure your deployment sever; you also do for your test environment.** This has caused a lot of issues in the past. For example, students would setup test environments on their personal laptops, but the environment would not match what the deployment server had. If your deployment and test environment differ too much, then deployment becomes a hassle especially running the app on the deployment server.

For example, in of the recent semesters the deployment server was running Node Js with NGINX webserver in front of the Node apps. Many students would have test environments on their PCs mimicking their node app on their PC but forgot to add NGINX as well. Forgetting NGINX caused the following problems:

- Apps would not even start after being pulled from GitHub
- Apps paths did not work
- Apps routes did not work.
- Apps were missing dependencies obtained from test environment.
- Apps would perform differently compared to test environment.
- And the list goes on.

A team from a previous semester had set up a test environment on a virtual machine that we gave to one of their teammates for another class. That teammate configured the VM to match their production server and then gave it to their team so they can use. It was not a requirement, but it helps ensure your test environment is as close to your production as possible. The tricky part is managing architecture changes and pushing these changes to each individual VMs.

NOTE about using deployment server: most choices will be on some form of a cloud service. To use cloud service properly one must understand how they charge. For example, it may happen that unintentionally students create many instances of SW or applications and have them running silently (unless they are stopped), thus incurring extra costs. It is imperative to avoid this, and we strongly suggest that somebody in the team be in charge of monitoring cloud usage. Also, cloud storage is very expensive so make sure you use it minimally and understand how it is charged (class project requires only minimal storage). Keep in mind that if your team uses app engine for google or the AWS equivalent stopping an instance is NOT enough. Killing an instance will only cause it to respawn since the app is active. To stop this the APP itself needs to be stopped or deleted which then will terminate all instances associated with that APP. Failure to do this may incur large costs for the service

We strongly recommend that you have one team member check cloud billing regularly.

4. Submission and Grading of M0

4.1 Submission of M0 for grading

All projects will be inspected and graded on the due date. The key for grading will be your team's correct installation and configuration of software stack, correct setup and usage of GitHub, and the deployment of your About Me Web Application from ***YOUR CHOICE OF REMOTE SERVER*** **(section 2).** Emphasis for M0 grading is on correctness (not so much on UI design of the final team page) and proper usage of development tools and deployment method.

Once you are done and ready for grading the team lead must send e-mail to class instructor with the following subject line and email body (**you MUST follow this protocol**). AND EACH TEAM MEMBER needs to record a video demonstrating that their systems and their individual accounts are setup in their computers (Local Laptop with IDE>>Cloud server>>GitHub), and upload it into Canvas.

Team lead E-mail subject line:
"CSC 648-848 Spring 2026 M0 Section N Team M"
(N is section 02; M is team number)

E-mail body:
In the e-mail say that M0 is ready for grading. Provide a link to your application and GitHub repository. Please do not send emails with no text and only a link or with empty body messages.

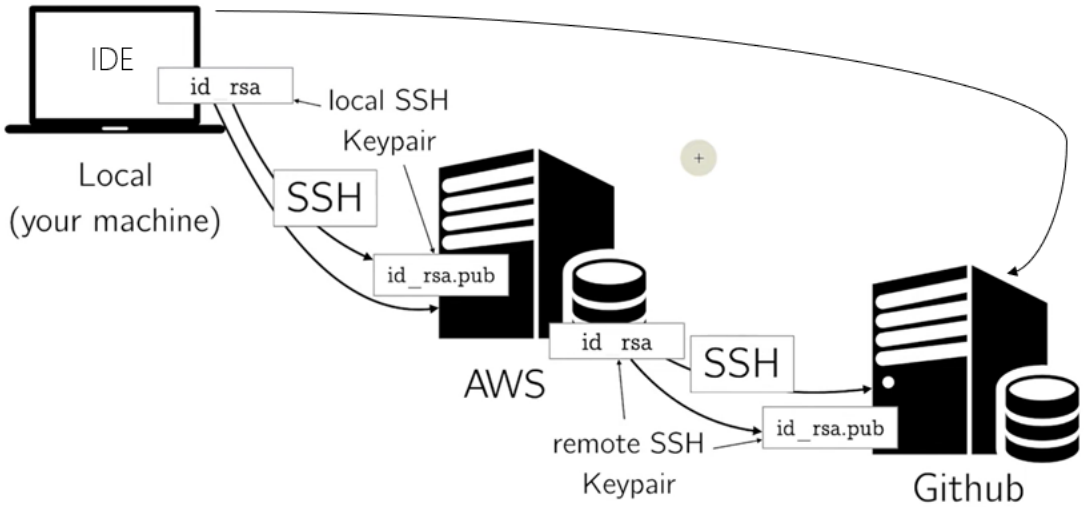
You must deliver M0 on schedule. Any delays must be approved by class CTO and asked for via e-mail at latest 24 h before the M0 deadline.

4.2 Grading of M0

This assignment is worth 10% of your class grade, as indicated in the syllabus. The grade for this assignment will be determined according to the following criteria:

Category	Points
Correctly Installed and Configured Software Stack	2
Correctly Populated Credentials Folder in GitHub & correct use of GitHub	3
Correct team WWW page functionality, deployment and proper usage of team's Software Stack for creating web page	3
Individual SE Dev Environment Video uploaded in Canvas	2
Total:	10

The individual SE Dev Environment Video should demonstrate the below connection in the student system/computer.


IDE
id_rsa
local SSH
Keypair
Local
(your machine)
SSH
id_rsa.pub
AWS
id_rsa
SSH
id_rsa.pub
remote SSH
Keypair
Github

Appendix

i. Sample Email for Software Stack Approval

Hello Class CEO and CTO

Below is a list of the technologies used in TeamNN's software stack:
Sever Host: Google Compute Engine 1vCPU 2 GB RAM
Operating System: Ubuntu 16.04 Server
Database: MySQL v……
Web Server: NGINX 1.12.2
Server-Side Language: Python
Additional Technologies: Web Framework: Flask
IDE: PyCharm InteliJ,
Web Analytics: Google Analytics
SSL Cert: Lets Encrypt (Cert Bot)
SASS: 3.5.5

Member's Familiarity with Server-Side Language on a scale of 1 to 5,
with 5 being very familiar and 1 being never used it.

Jane: 5
Jack: 2
John: 3
Joe: 1
Jill: 4
Jake: 5

Then any extra information you think is necessary.

Best,
TeamNN

ii. Email Submission for Milestone 0

Hello Class CEO and CTO

Our team has completed Millstone 0 and is ready for grading.
1. Link to application
2. Link to GitHub Repo

Please state (say YES) that your team has correctly populated the credentials folder.

Then any extra information you think is necessary.
Best,
TeamNN

APPENDIX

Connecting to MySQL Database via cmd line or Workbench

Connecting to MySQL Server via MySQL Workbench

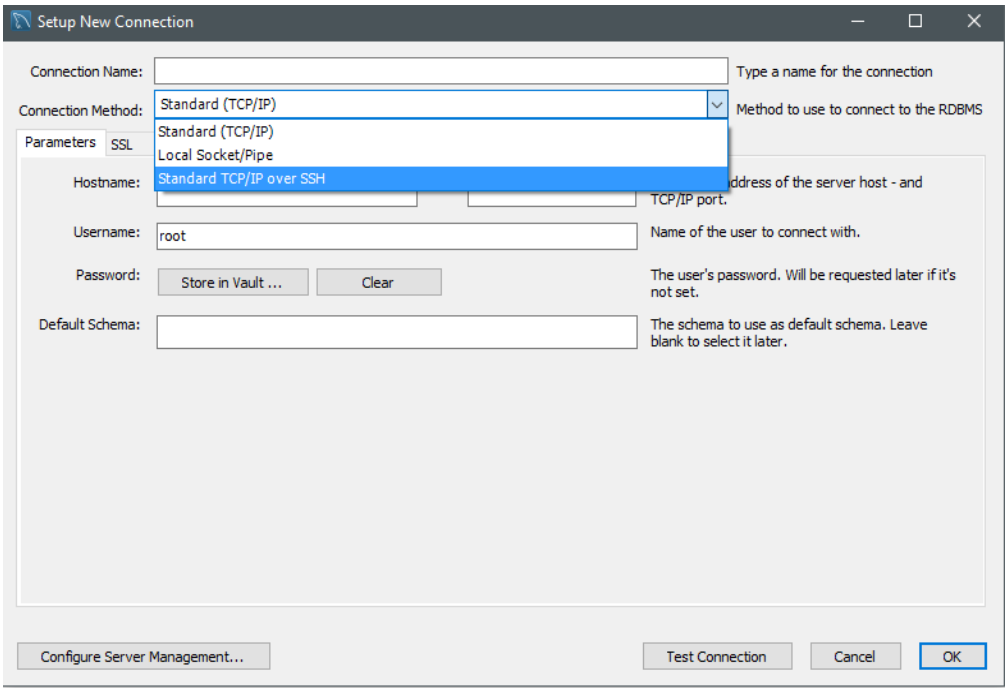
There are two ways you may access your database. One way is logging into the remote Google Compute Engine server and accessing your database via command line with the following commands:

mysql -u username -p
(where username is your database username and password is your db password)
connect db_name
(where db name is the database name you wish to connect to)

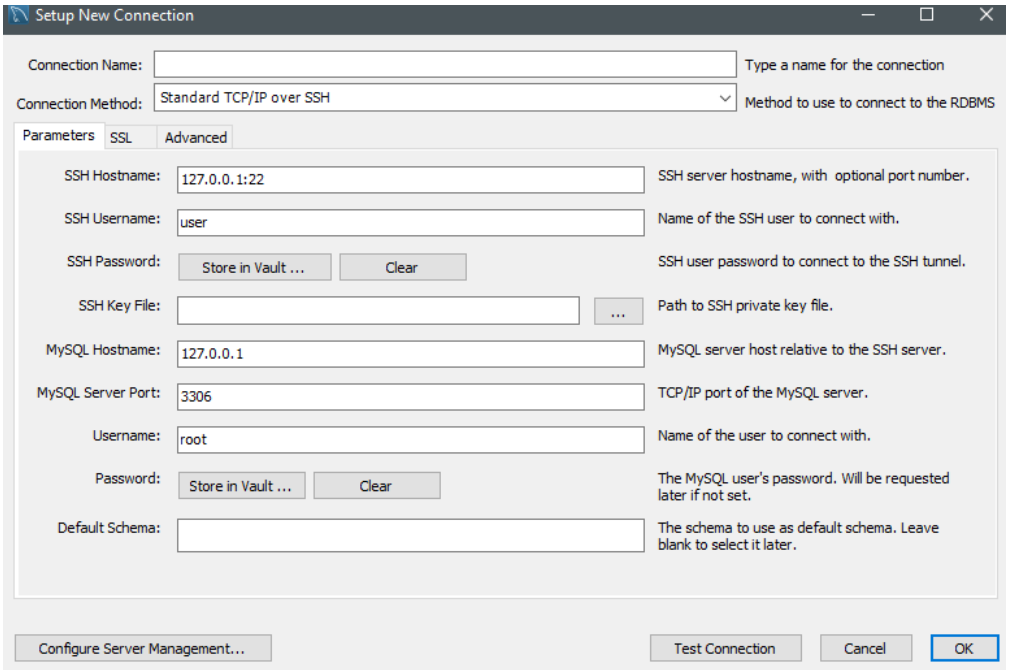
At this point you may start executing SQL commands against your connected database.

The other way is via MySQL Workbench. This is GUI application used to manage MySQL databases.

To connect to your database, you first need to create a new connection. This can be done by clicking the plus sign (+) near the text MySQL Connections. This will open a new tab. You will then select a connection type. The connection to select is "Standard TCP/IP over SSH". Figure shown below.



Next, you will need to fill in the required information for creating a new connection over SSH.



Using the above figure as an example, fill in the following information:

- Connection Name: teams_db
- SSH Hostname: your host name (could be an IP)
- SSH username: your_ssh_username
- SSH Key File: path you your key file
- MySQL Hostname: 127.0.0.1
- MySQL Server Port: 3306
- Username: database username, same as shell account name
- Password: database password, should be student id, unless changed.

Default Schema: student_username, this your database name, where username is your database account name.

Other Option is to Connect with username and passed.

This is a simple process and is the default way to connect. Therefore, the steps will not be shown. But you should note the following:

- To connect to database remotely not via SSH a port needs to be opened on the server pointed to the database. Note that this is against good security practices.
- A user account in your database needs to be made for the connection. DO NOT use root to do this, while easy and simply its lazy and goes against good security practices.